

Solar O appendix

Sun · O · downloadable appendix

Solar oxygen: 8.643 ± 0.076

Asplund, Amarsi & Grevesse 2021: 8.69 ± 0.04 · this measurement -0.047 dex

IAG FTS · red-optical · Reference Grade · SET-AGSS21 · ENGINE-A-3DNLTE · n = 6. The top-ranked admitted product (Reference Grade, then the most complete treatment, then line count, then total uncertainty); independent products and engines are not averaged together.

Highlighted Products

VIS

8.810 ± 0.421

±0.421 stat · ±0.006 syst

FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied · solar_harps_molecfits_corrected · Codex Grade

red-optical

8.643 ± 0.076

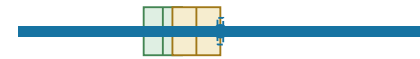
±0.048 stat · ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied · solar_iag · Reference Grade

Error-bar Forest Plot

VIS — harps solar_harps_molecfits_corrected A(O) dex; axis 8.36-9.26

FORB-OI_6300 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Codex Grade · ALL · n=1



8.810
±0.421 stat ±0.006 syst

VIS axis: 8.36 · 8.58 · 8.81 · 9.04 · 9.26 A(O) dex

red-optical — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.48-9.00

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Reference Grade · ALL · n=6



8.803
±0.053 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=6



8.643
±0.048 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann · SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=6



8.674
±0.042 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-LTE · SYNTH · 1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.826
±0.132 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE · Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not supplied
Codex Grade · ALL · n=2



8.665
±0.121 stat ±0.043 syst

red-optical — iag_fts_solar_atlas solar_iag A(O) dex; axis 8.48-9.00

indicator not supplied · O I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2



8.694
±0.107 stat ±0.043 syst

red-optical — kpno_solar_atlas solar_kpno_kurucz2005_corrected A(O) dex; axis 8.48-9.00

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=6



8.805
±0.053 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=6



8.644
±0.049 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=6



8.675
±0.043 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.826
±0.133 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=2



8.666
±0.122 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2



8.695
±0.107 stat ±0.043 syst

red-optical — kpno_solar_atlas solar_kpno_molecfit_corrected A(O) dex; axis 8.48-9.00

SET-AGSS21 · O I · Synth · 1D-LTE · SYNTH · 1D-LTE ·
line set not supplied
Reference Grade · ALL · n=6



8.811
±0.059 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 3D-NLTE · Amarsi · SYNTH
· ENGINE-A-3DNLTE · line set not supplied
Reference Grade · ALL · n=6



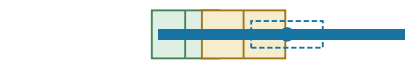
8.651
±0.055 stat ±0.043 syst

SET-AGSS21 · O I · Synth · 1D-NLTE · Bergemann ·
SYNTH · ENGINE-A · line set not supplied
Reference Grade · ALL · n=6



8.681
±0.048 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-LTE · SYNTH ·
1D-LTE · line set not supplied
Codex Grade · ALL · n=2



8.812
±0.154 stat ±0.043 syst

indicator not supplied · O I · Synth · 3D-NLTE ·
Amarsi · SYNTH · ENGINE-A-3DNLTE · line set not
supplied
Codex Grade · ALL · n=2



8.651
±0.143 stat ±0.043 syst

indicator not supplied · O I · Synth · 1D-NLTE ·
Bergemann · SYNTH · ENGINE-A · line set not
supplied
Codex Grade · ALL · n=2



8.682
±0.127 stat ±0.043 syst

red-optical axis: 8.48 · 8.61 · 8.74 · 8.87 · 9.00 A(O) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Solid bars show the supplied

statistical uncertainty; wireframes show the supplied systematic uncertainty. The statistical basis and any incomplete error budget remain attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

How this was measured. One row per band. The indicator set is what was fitted; the holding is the solar spectrum it was fitted against; the engine and treatment say how. Dominant term is the largest single contributor to the reported uncertainty, and gf rung is the pedigree of the oscillator strengths that pool could reach.

Solar oxygen context

Oxygen is traced by forbidden [O I], permitted O I and molecular OH indicators where available. Blends, departures from LTE and atmospheric structure affect these families differently, so their results retain separate identities.

[C/O indicator strategy](#) · [Methodology](#)

Open provenance decisions

O I 615.8nm: □ DISAGREES — reported to Ryan, not resolved. . Store log gf -0.409 (); summed pull -0.2966 (B+).

O I 630.0nm: □ DISAGREES — reported to Ryan, not resolved. □ store claims A, pull publishes B+. Store log gf -9.717 (A); summed pull -9.7744 (B+).

OPEN. Tachiev and NIST differ systematically by -0.016 dex on the 777 triplet and -0.005 dex on 8446. AGSS21 names Tachiev for N i only and states no O i gf source. Both values are reported in oi_tachiev_vs_nist.csv; NIST is what the store already carried and this script does not change it. Ryan decides.

[Adjudication record](#) · [Indicator gf checks](#)

Download the evidence

[Download PDF](#) — Solar O appendix

[Source feed](#) — includes quarantined fit outputs, not adopted abundances

[Complete product visibility audit](#)

[Shared website/PDF report](#)

[Reproducibility manifest](#)

[Complete references and product-to-source bindings](#)

Reproducibility

Generated from the pinned CNO campaign snapshot. Source commit

886a0a3df4be16f56e3cf3e6ff3220c88b99a38

Feed O.json v1.27 · updated 2026-10-01T21:07:09Z

Generated 2026-10-01T15:12:46-06:00 (source commit timestamp)

Feed SHA-256 95212061e3a472ed2d2c9e726badbc094fe8c2774c67d26b02b75a508a31fe5e

Generator scripts/generate_cno_publication.py v1.0.0

[Pinned scientific source](#)

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar O atomic/molecular indicator lineage and atmospheric sensitivity; comparison literature, not an adopted Codex abundance. Sources: [\[1\]](#), [\[2\]](#).

Atomic data and gf authorities. The recorded O I/O II base pedigree is Wiese-Fuhr-Deters / Opacity Project theory. Per-transition exceptions and the open O I source decisions remain explicit. Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Atomic data and gf authorities. O I triplet source record: NIST (this script) — Tachiev NOT adopted; OPEN — referred to Ryan (RYA-1214 EOS) Sources: [\[3\]](#), [\[6\]](#).

Model and grid sources. Registered O I line/grid lineage: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — line-by-line 3D-NLTE grids built on the Amarsi+2018 O I model atom. The current feed treatment remains authoritative; a grid reference does not imply a correction was applied. Sources: [\[7\]](#), [\[8\]](#).

Atomic data and gf authorities. Atomic O I entry; source-family pedigree only, not a substitute for a per-transition gf join. Sources: [\[3\]](#), [\[4\]](#), [\[5\]](#).

Model and grid sources. Feed treatment: 1D-LTE; no NLTE/3D correction is claimed for this entry. Turbospectrum is the synthesis implementation. Sources: [\[9\]](#).

Special diagnostics. The synthesis sidecar names Johansson's Ni I blend data, Storey & Zeippen's forbidden-line gf lineage and Caffau's CO5BOLD comparison anchor. The feed entry remains 1D-LTE and quarantined; the sidecar's cited anchor is not promoted to a measured abundance. Sources: [\[10\]](#), [\[11\]](#), [\[12\]](#).

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 3D non-LTE - 1D LTE on STAGGER (table2 C I / table3 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 A) Sources: [\[7\]](#), [\[8\]](#).

Model and grid sources. Registered C/O model lineage; per-product grid provenance must confirm the applied treatment. Run provenance: Amarsi, Nissen & Skuladottir 2019, A&A 630, A104 — 1D non-LTE - 1D LTE (table5 C I / table6 O I); PER-LINE additive correction, grid lines only (match ≤ 0.05 A) Sources: [\[7\]](#), [\[8\]](#).

Product-to-source index

O-products-0ba7067733f9 · products · VIS · solar_harps_molecfit_corrected · FORB-OI_6300 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#), [\[10\]](#), [\[11\]](#), [\[12\]](#)

O-products-d00ae41a9b00 · products · red-optical · solar_iag · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-a911fb18ea46 · products · red-optical · solar_iag · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-bbeb69b508ea · products · red-optical · solar_iag · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-c4bdfbe8ac19 · products · red-optical · solar_iag · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-f02b3eb1b0e5 · products · red-optical · solar_iag · not supplied · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-1a72432a141c · products · red-optical · solar_iag · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-711c33d498f6 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-297d2f9df123 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-24673d0fe3f0 · products · red-optical · solar_kpno_kurucz2005_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-4961ac96e190 · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-e0acb098043b · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-294d60b5a66b · products · red-optical · solar_kpno_kurucz2005_corrected · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-1d0c52443ad4 · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-84774ef30efe · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-faf80741533b · products · red-optical · solar_kpno_molecfit_corrected · SET-AGSS21 · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-e7b874120b2b · products · red-optical · solar_kpno_molecfit_corrected · not supplied · 1D-LTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[9\]](#)

O-products-093a3dc3a2d6 · products · red-optical · solar_kpno_molecfit_corrected · not supplied · ENGINE-A-3DNLTE · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

O-products-b0c45bc56d31 · products · red-optical · solar_kpno_molecfit_corrected · not supplied · ENGINE-A · not supplied · [\[1\]](#), [\[3\]](#), [\[4\]](#), [\[5\]](#), [\[7\]](#), [\[8\]](#)

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. *A&A* 653, A141 DOI: 10.1051/0004-6361/202140445. [source record](#)

[2] — Amarsi et al. 2021, *A&A* 656 A113 DOI: 10.1051/0004-6361/202141384. [source record](#)

Atomic data and gf authorities

[3] — Kramida, A.; Ralchenko, Yu.; Reader, J.; NIST ASD Team (2024). NIST Atomic Spectra Database (version 5.11/5.12). National Institute of Standards and Technology, Gaithersburg, MD DOI: 10.18434/T4W30F. [source record](#)

[4] — Wiese, W. L.; Fuhr, J. R. (2006). New Critical Compilations of Atomic Transition Probabilities for Neutral and Singly Ionized Carbon, Nitrogen, and Iron. Proceedings of the NASA Laboratory Astrophysics Workshop (14-16 February 2006, UNLV, Las Vegas) [source record](#)

[5] — Wiese, W. L.; Fuhr, J. R.; Deters, T. M. (1996). Atomic Transition Probabilities of Carbon, Nitrogen, and Oxygen: A Critical Data Compilation. JPCRD Monograph 7. [source record](#)

[6] — Tachiev & Froese Fischer 2002, *A&A* 385, 716 DOI: 10.1051/0004-6361:20011816. [source record](#)

Model and grid sources

[7] — Amarsi, A. M.; Nissen, P. E.; Skuladottir, A. (2019). Carbon, oxygen, and iron abundances in disk and halo stars. *A&A* 630, A104 DOI: 10.1051/0004-6361/201936265. [source record](#)

[8] — Amarsi, Barklem, Asplund, Collet & Zatsarinny 2018, *A&A* 616, A89 DOI: 10.1051/0004-6361/201832770. [source record](#)

[9] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. *A&A* 669, A43 DOI: 10.1051/0004-6361/202243673. [source record](#)

Special diagnostics

[10] — Johansson, S.; Litzen, U.; Lundberg, H.; Zhang, Z. (2003). Experimental f-Value and Isotopic Structure for the Ni I Line Blended with [O I] at 6300 Å. *ApJ* 584, L107 DOI: 10.1086/374037. [source record](#)

[11] — Storey, P. J.; Zeippen, C. J. (2000). Theoretical values for the [O III] 5007/4959 line-intensity ratio and homologous cases. *MNRAS* 312, 813 DOI: 10.1046/j.1365-8711.2000.03184.x. [source record](#)

[12] — Caffau, E.; Ludwig, H.-G.; Steffen, M.; Livingston, W.; Bonifacio, P.; Malherbe, J.-M.; Doerr, H.-P.; Schmidt, W. (2015). The photospheric solar oxygen project. III. Investigation of the centre-to-limb variation of the 630 nm [O I]-Ni I blend. *A&A* 579, A88. [source record](#)